ADVANCED INTELLIGENT DISCOVERY
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CALL FOR PAPERS / SPECIAL ISSUE

AI-Aided Discovery in Archaeology: Applications of ML/AI to Archaeological Data Collection, Analysis, and Interpretation

Advanced Intelligent Discovery

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Overview

Artificial intelligence (AI) and machine learning (ML), from convolutional neural networks and LiDAR-based deep learning to large language models (LLMs), are transforming how archaeological sites are discovered, how artifacts are classified, and how landscapes and settlement patterns are interpreted.

The field is advancing unevenly. Remote-sensing detection is well established, while newer areas such as LLM-assisted interpretation remain thinly explored. Data and computational access, publishing costs, and reproducibility still limit who can take part and how far results can be trusted and reused.

This Special Issue of Advanced Intelligent Discovery invites contributions spanning the full pipeline, from raw remote-sensing and archaeometric data to interpretation and practice.

Topics of Interest

Submissions may address, but are not limited to:

AI-assisted remote sensing and site detection

- CNN, deep-learning, and hybrid ML pipelines
- LiDAR, SAR, multispectral satellite, historical aerial, and drone imagery
- Detection of mounds, hillforts, field systems, tumuli, qanats, and other subsurface or landscape-scale features
- Applications from Scandinavian agricultural landscapes to the Sahara and the Pacific islands

The Special Issue particularly welcomes methodological novelty over further detection-application case studies.

Automated artifact and material classification

- Computer-vision and ML approaches to typology
- Segmentation and classification of ceramics and other material culture
- Self-organizing-map and elemental-analysis-linked classifiers

Large language and vision-language models in archaeological interpretation

- LLMs and vision-language models (VLMs) as interpretive aids and natural-language interfaces
- Magnetometer-data interpretation
- Artifact-typology assistance

This is one of the thinnest and fastest-moving areas of development, and one this Special Issue particularly wants to grow.

Generative AI in archaeological reconstruction

- Generative models for artifact and material reconstruction
- Architectural and site reconstruction
- Validating, documenting, and communicating uncertainty in generated reconstructions

Predictive and spatial modelling

- ML- and AI-based predictive modelling of site location and settlement patterns
- Land-use and landscape evolution
- Encoder and forecasting models for heritage risk
- Looting and site-protection monitoring

Archaeometric and provenance-focused AI applications

- ML applied to material and biomolecular archaeometry
- Provenance and sourcing
- Chronology
- Bridging archaeological science and computational methods

AI and archaeological theory

- Theoretical frameworks for AI-aided archaeological inference
- How AI reshapes interpretation, knowledge production, and disciplinary practice
- Relating machine-learned patterns to archaeological theory

Critical perspectives, reviews, and infrastructure

- Systematic reviews and landscape assessments
- Critical and ethical perspectives on AI in archaeological practice
- Equity, ethics, and access
- Methodological challenges and reproducibility

Article Types

The Special Issue welcomes:

- Original Research Articles
- Review Articles
- Perspectives and critical commentaries

Important Dates

MILESTONE	DATE
Call announced	**30 November 2026**
Submission deadline	**31 May 2027**
Publication	**Online upon acceptance**

Why Contribute?

This Special Issue aims to map where AI and ML are changing archaeological practice, and where they still fall short.

Particular emphasis will be placed on contributions that address:

- Methodological novelty beyond detection case studies
- Large language and vision-language models in archaeological interpretation
- Reproducibility, ethics, and equity in AI-aided archaeology
- Barriers to broader adoption: data and computational access, open-access publishing costs, and reproducibility
- Work from regions currently underrepresented in this literature, including Sub-Saharan Africa, South and Southeast Asia, and Latin America

We welcome contributions from academic, heritage-agency, and cultural-institution groups worldwide.

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How to Submit

Manuscripts should be submitted through **https://authors.wiley.com/journal/AIDI**, selecting this Special Issue. Prospective authors are welcome to send a short title and abstract to the Guest Editors beforehand to check fit.

Advanced Intelligent Discovery

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